

LUKMAN HAKIM (he/him)

Geospatial Professional | M.Sc. Photogrammetry and Geoinformatics

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PROFILE

Geospatial professional with 6+ years of experience in GIS, remote sensing, spatial modelling, and geodata management for environmental, conservation, and public-sector geospatial projects. Skilled in QGIS, ArcGIS Pro, Python/GeoPandas, PostGIS, spatial data quality assurance, and reproducible geodata workflows. Experienced in integrating heterogeneous datasets, producing planning-relevant spatial analyses, preparing decision-support maps, and communicating GIS outputs to interdisciplinary stakeholders. Highly motivated to apply geospatial analysis to renewable-energy project development, land-potential assessment, and climate-focused infrastructure planning.

CORE COMPETENCIES

Spatial Databases & Data Management

SQL · PostgreSQL/PostGIS · Geodata management · Data integration · Data quality assurance · Metadata and workflow documentation

Programming & Automation

Python · TensorFlow · Pandas · PyTorch · API-based data processing · Automation · Reproducible workflows

WebGIS & Geospatial Services

GeoServer · GeoDjango · Leaflet · Flask · JavaScript · JSON · REST APIs · OGC services

GIS & Spatial Analysis

QGIS · ArcGIS Pro · Trimble · Cartography · Coordinate reference systems · Conservation planning · Spatial prioritization · Zoning support · Environmental monitoring · Multi-source spatial datasets

Remote Sensing & Spatial Modelling

Aerial Imagery · Satellite Imagery · Google Earth Engine · ENVI · SNAP · Image classification · Machine learning

Tools & Collaboration

Git · Docker · FME · Technical documentation · Interdisciplinary teamwork · Stakeholder communication · Capacity Building

PROFESSIONAL EXPERIENCE

Landesamt für Geoinformation und Landentwicklung Baden-Württemberg — Student Job

(06/2025 – 11/2025)

Identification of green roof areas in Baden-Württemberg using LoD2 building data and PlanetScope imagery

- Processed LoD2 building geometry (slope, aspect, area, height), semantic building attributes, and PlanetScope satellite imagery to support a 2025 roof-plane-level green roof inventory for Baden-Württemberg.
- Developed Python/GeoPandas workflows for geodata preparation, feature extraction, quality control, and machine-learning-ready spatial datasets.
- Worked with large-scale official geodata in a public-sector context, including structured spatial data handling, documentation, and reproducible processing.

The Nature Conservancy Indonesia — GIS & Remote Sensing Specialist

(07/2019 – 08/2024)

- Managed, processed, and integrated heterogeneous geospatial datasets from satellite imagery, field data, participatory mapping, administrative sources, and conservation planning workflows.
- Developed GIS and remote-sensing workflows for mangrove mapping, benthic habitat classification, herpeto-aves-mammals occurrence, and conservation decision-support ([contribution preview](#))
- Applied multi-temporal satellite data, including Landsat, Sentinel, and Planet NICFI, within Google Earth Engine for environmental monitoring and spatial modelling.
- Supported the design and optimization of Marine Protected Area networks using MARXAN, Gap Analysis, and spatial prioritization methods for local, national, and regional zoning ([contribution preview](#))
- Prepared thematic maps, spatial datasets, technical outputs, documentation, and capacity building for interdisciplinary projects involving NGOs, government agencies, researchers, and local communities ([contribution preview](#))
- Mentored junior staff in GIS, remote sensing, data preparation, and spatial analysis workflows.

Research Institute for Tuna Fisheries Bali — Internship

(08/2018 – 09/2018)

- Collected fisheries-related spatial and environmental data for bachelor thesis research.
- Applied remote sensing and Generalized Additive Model (GAM) methods for distribution mapping of Yellowfin Tuna (*Thunnus albacares*) in the Indian Ocean.

Selected Early Experience

BAPPEDA Gunungkidul — Research and Survey Team

(05/2017 – 12/2017)

- Site Selection for D.I. Yogyakarta Planetarium & Observatory, Gunung Kidul Regency
- Supported GIS-based site selection and field survey activities for a regional planning project.
- Assisted in collecting, checking, and interpreting spatial and field data to evaluate potential locations.
- Contributed to planning-oriented geospatial analysis in collaboration with a local government planning agency.

EDUCATION

[Master of Science \(M.Sc.\), Photogrammetry and Geoinformatics — Hochschule für Technik Stuttgart](#)

(2024 – 2026, DAAD Scholarship Awardee)

[Bachelor of Science \(B.Sc.\), Cartography and Remote Sensing — Universitas Gadjah Mada](#)

(2014 – 2019, BIDIKMISI Scholarship Awardee)

LANGUAGES

Indonesian: Native | English: C1 | German: B1-B2

SELECTED PUBLICATIONS

- Fajariyanto, Y., Green, A., **Hakim, L.**, et al. (2024). [A Resilient Marine Protected Area Network Design: A First for the Arafura–Timor Seas. Coastal Management](#), 52(3), 147–173.
- Prakoso, D. A., **Hakim, L.**, et al. (2023). [The Dynamics of Mangrove and Pond Changes in East Kalimantan, Indonesia. IOP Conference Series: Earth and Environmental Science](#).
- **Hakim, L.**, Lazuardi, W., Astuty, I. S., et al. [Assessing WorldView-2 Satellite Imagery Accuracy for Bathymetry Mapping in Pahawang Island, Indonesia](#). IOP Conference Series: Earth and Environmental Science.
- Aung Kyaw, H., Ayala Rico, J., Awontemi, I. A., **et al.** (2026). Urban Vegetation Mapping from High-Resolution Aerial Imagery with Deep Learning. DGPF Jahrestagung 2026. Manuscript in submission.

ADDITIONAL QUALIFICATIONS

Small UAS Remote Pilot · Open Water Diver

EXTRACURRICULAR

2023 Presenter at 5th Asia Pacific Coral Reef Symposium, NUS Singapore

2017 Presenter at World Blue Carbon Conference, Jakarta Convention Center Indonesia

2017 Presenter at Seminar Nasional Geomatika, Geospatial Information Agency Indonesia

Additional early experience includes serving as Vice President of the Geography Study Club at Universitas Gadjah Mada, and winning national college level debate competitions.

REFERENCE

- Prof. Dr. rer. nat. Michael Mommert | Hochschule für Technik Stuttgart | michael.mommert@hft-stuttgart.de
- Maryse Hilemann | Landesamt für Geoinformation und Landentwicklung Baden-Württemberg | Maryse.Hilemann@lgl.bwl.de
- Yusuf Fajariyanto | The Nature Conservancy Indonesia | yfajariyanto@tnc.org